

ANTHONY KANG

Chicago, IL ♦ Contact via this site ♦ GitHub: ping48

EDUCATION

Harvey Mudd College (HMC)

Bachelor of Science, Engineering

May 2024

Relevant Coursework: Microprocessors, Computer Architecture and Digital Design, Data Structures and Algorithms, Introduction to Machine Learning, Systems Engineering, Radio Frequency Circuits, Mechanical Design

SKILLS

Languages: Proficient in C/C++, Python; Acquainted with SystemVerilog, Java, SQL, Javascript; Fluent in Korean

Software and Systems: Git, Linux, Docker, CubeMX, Oscilloscopes, Spectrum Analyzer, CAN, SPI, UART, I2C

WORK EXPERIENCE

Ascend Engineering – Embedded Engineer, Chicago, IL

Sept. 2025 – Present

- Developed a multi-camera live video streaming system in C/C++ for a long-range drone, delivering both an onboard FPV view and high-resolution external payload camera with seamless in-flight switching between sources
- Built pipeline to ingest payload's encoded RTSP feed and copy raw frames into shared memory, signaling separate process to feed frames into hardware encoder — reused shared-memory hand-off logic to render object detection overlays onto outgoing streams
- Implemented adaptive bitrate control based on radio link quality — at worst conditions dropping one of the two streams entirely within 30 ms — keeping the operator's feed smooth as conditions degrade in flight
- Built network-health telemetry node for Neuron, a ROS 2 platform bridging a swarm robotics orchestrator to individual vehicles. Implemented primary backend using Linux netlink kernel API (pyroute2) for robustness
- Designed node with a network-interface agnostic backend layer, supporting Doodle radios and Wi-Fi mesh at launch, with clean extension points for future radio hardware as the platform grows

Doosan Bobcat – Embedded Software Engineer I, Fargo, ND

July 2024 – Sep 2025

- Developed vehicle localization system in C++ on embedded Linux, fusing IMU, GPS, and wheel velocity data through an Extended Kalman Filter
- Wrote drivers for underlying sensor stack — u-blox F9P/F9H GNSS, communication with IMU over I2C. Created NTRIP client and routed corrections to u-blox, delivering RTK fix and cm-level accuracy GPS data
- Built derating mechanism for geofencing application, stopping construction vehicle near hazardous zones. Used look-up tables of experimentally collected data to predict stopping distance based on engine RPM and speed
- Routed derate commands between hub and drive ECUs over J1939 CAN, controlling swash-plate positions

Doosan Bobcat – R&D Intern, Bismarck, ND

May 2023 – Aug. 2023

- Built foundation for autonomous construction vehicle by programming microcontrollers in C++, debugging with Vector CANalyzer, and integrating CANOpen-based wheel velocity sensors
- Enabled safe remote-control operation by translating commands into vehicle outputs, implementing PID control, and applying rate limiting to protect system health

PROJECTS

Radio Receiver Architecture - Claremont, CA

December 2023

- Designed and built a superheterodyne receiver to decode a secret message transmitted at ~2.3 GHz
- Created a link budget to calculate the noise temperature of the system and select a suitable amplifier from path loss

Roll Correcting Robot Cat (gabrielks.github.io/microps-cat-portfolio) - Claremont, CA

Oct. 2022 – Dec. 2022

- Designed and 3-D printed a robot that can correct its angular displacement (up to 180 degrees) by introducing angular momentum through a motor and machined flywheel
- Programmed an STM32L432KC microcontroller in C++ to read angle measurements from an IMU through USART and send the data to an iCE40 UltraPlus FPGA through SPI
- Programmed PID control in SystemVerilog (FPGA) for a target number of wheel turns and implemented a module to read quadrature signals from a position encoder to provide closed-loop feedback

Configuring Timer Registers on Microcontroller – Claremont, CA

Oct. 2022

- Programmed bare metal timer drivers on an STM32L432KC microcontroller in C++ for duration and PWM
- Played Für Elise and Darude Sandstorm by taking in an array of frequency (Hz) and beat length